

26 — PERCEPTRON USING SCIKIT-LEARN

Aim

To implement a Perceptron classifier using scikit-learn on a linearly separable binary classification dataset and evaluate its classification accuracy.

Algorithm + Procedure

1. Generate a synthetic binary classification dataset using `make_classification()`.
2. Create two input features and two target classes.
3. Split the dataset into training and testing sets.
4. Create a Perceptron classifier using scikit-learn.
5. Set the maximum number of iterations and learning rate.
6. Train the Perceptron using the training dataset.
7. Predict the class labels for the test dataset.
8. Compare the predicted values with the actual test values.
9. Calculate the classification accuracy.
10. Display the accuracy of the trained Perceptron classifier.

Code

```
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score

# Generate a binary classification dataset
X, y = make_classification(
    n_samples=200,
    n_features=2,
    n_redundant=0,
    n_clusters_per_class=1,
    class_sep=2.0,
    random_state=1
)
```

```
# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=1
)

# Create and train Perceptron
model = Perceptron(
    max_iter=1000,
    eta0=0.1,
    random_state=1
)

model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
```

Output

Accuracy: 0.9833333333333333

Result

The scikit-learn Perceptron classifier was successfully implemented and trained on a synthetic binary classification dataset. The model achieved high classification accuracy on the test dataset.